Lab 4: Unit-root tests and ARIMA modelling

ADF, KPSS, differencing, and ARIMA fitting on real data

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Goals

- Distinguish a trend-stationary (TS) process from a difference-stationary (DS) process via simulation.
- Run the Augmented Dickey–Fuller (ADF) and KPSS tests, read their p-values, and combine them.
- Apply differencing to remove a unit root and verify the transformation by re-running the tests.
- Fit an ARIMA(p, d, q) model on a non-stationary series and produce a short forecast.

Setup

- Python 3 with `numpy`, `matplotlib`, `statsmodels`.
- `numpy.random.seed(42)`.
- Dataset: `data/SouvenirSales.csv` (Australian souvenir shop, monthly, Jan 1995 – Dec 2001, $T = 84$).

Warning

`adfuller(y)` returns the tuple `(stat, pvalue, usedlag, nobs, critvalues, icbest)`. `kpss(y, regression="c")` returns `(stat, pvalue, lags, critvalues)` (with a FutureWarning about p-value clamping that you can ignore).

Exercise 1 – TS vs. DS by simulation

We compare two visually similar series.
(1) Simulate $T = 200$ observations of a trend-stationary process

$$Y_t = 0.05\,t + \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1).$$

(2) Simulate $T = 200$ observations of a random walk with drift:

$$Z_t = Z_{t-1} + 0.05 + \eta_t, \qquad \eta_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad Z_0 = 0.$$

(3) Plot Y_t and Z_t on the same figure. Visually, can you tell which is TS and which is DS?
(4) Detrend each series by OLS regression on t and plot the residuals. Comment: the TS residual should look like white noise; the DS residual still wanders.

Exercise 2 – ADF test

(1) Run `adfuller` on Y_t and on Z_t from Exercise 1. Report for each series:

- the ADF statistic $t_{\widehat{\gamma}}$;
- the 1%, 5%, 10% critical values;
- the p-value;
- the conclusion at the 5% level (reject H_0: unit root, or fail to reject).

(2) On the random walk with drift Z_t, the ADF test should *fail to reject* the unit-root null. On the trend-stationary Y_t, you must use `regression="ct"` (constant + trend); otherwise the deterministic trend would inflate the test statistic.
(3) Repeat the ADF test on $\Delta Z_t = Z_t - Z_{t-1}$. You should now **reject** the unit root: differencing has made the series stationary.

Exercise 3 – KPSS test

The KPSS test reverses the null and the alternative: H_0 is stationarity, H_1 is unit root.
(1) Run `kpss(y, regression="c")` on Y_t (after removing the linear trend, use `regression="ct"` on the raw series) and on Z_t. Report KPSS statistic, p-value, conclusion.
(2) Build the ADF/KPSS combination matrix on the real series of Exercise 5: which of the four cells (stationary, trend-stationary, difference-stationary, ambiguous) does it land in?

Exercise 4 – Differencing the random walk

(1) Compute $\Delta Z_t = Z_t - Z_{t-1}$ on the simulated random walk of Exercise 1. Plot ΔZ_t vs. t.
(2) Plot the sample ACF of Z_t and of ΔZ_t up to lag 20. Compare: the ACF of Z_t decays slowly (signature of a unit root); the ACF of ΔZ_t should be near zero for all $h \geq 1$.
(3) Run `adfuller` and `kpss` on ΔZ_t. Confirm that one differencing was enough.

Exercise 5 – ARIMA on Souvenir sales

(1) Load `data/SouvenirSales.csv` and take logs: $y_t = \log(\text{sales}_t)$. Plot y_t.
(2) Run ADF and KPSS on y_t (with `regression="ct"` since there is a clear linear trend on the log scale). Conclude.
(3) Fit the model

$$\text{SARIMA}(0,1,1)(0,1,1)_{12}$$

on y_t using `statsmodels.tsa.statespace.sarimax.SARIMAX`. Report the estimated coefficients and the AIC.
(4) Plot the residuals of the fitted model. Run a Ljung–Box test at lag 24 on the residuals: are they white noise?
(5) Forecast the next 12 months. Plot the in-sample fit and the forecast with 95% confidence intervals.

Warning

The KPSS test in `statsmodels` returns a p-value clamped to $[0.01, 0.10]$. To get a finer reading, use the printed critical values (1%, 2.5%, 5%, 10%) directly.